Mellinger-Garrett_individual-assignment-05

Garrett Mellinger

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")
```

```r
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
                                "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning `sites` object

Goal: create a new object from `sites` that only includes NCOS sites

Use: filter downstream data frames using new object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

Goal: convert character strings in existing `taxon_list` data frame into snake case

Use: join with aquatic inverts to attach taxonomic information to species IDs

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning `water_quality`

Goal: create data frame with mean pH, mean DO (in mg/L), and mean salinity (in ppt) for each date and site

Use: join with aquatic inverts to attach water quality information to surveys

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning `aq_ins`

Goal: create long format data frame in which each row contains the counted individuals in each taxon during each survey on each date with all the taxonomic information for the taxon and the water quality information for the survey

Use: visualize relationships between water quality parameters and aquatic invertebrate abundances

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
```

```
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter
# (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~  "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median",
                               .na_rm = TRUE),
       site_full = fct_rev(site_full))
```

1. Visualizing differences across sites in major taxon abundance

a. Planning

- x-axis: site
- y-axis: log-transformed ostracod average abundance/L
- geometry to represent average abundance/L: points representing individual log-transformed average ostracod abundance/L values
- geometry to represent medians: larger points with a different color, shape, and size representing the median abundance/L at each site

b. Wrangling

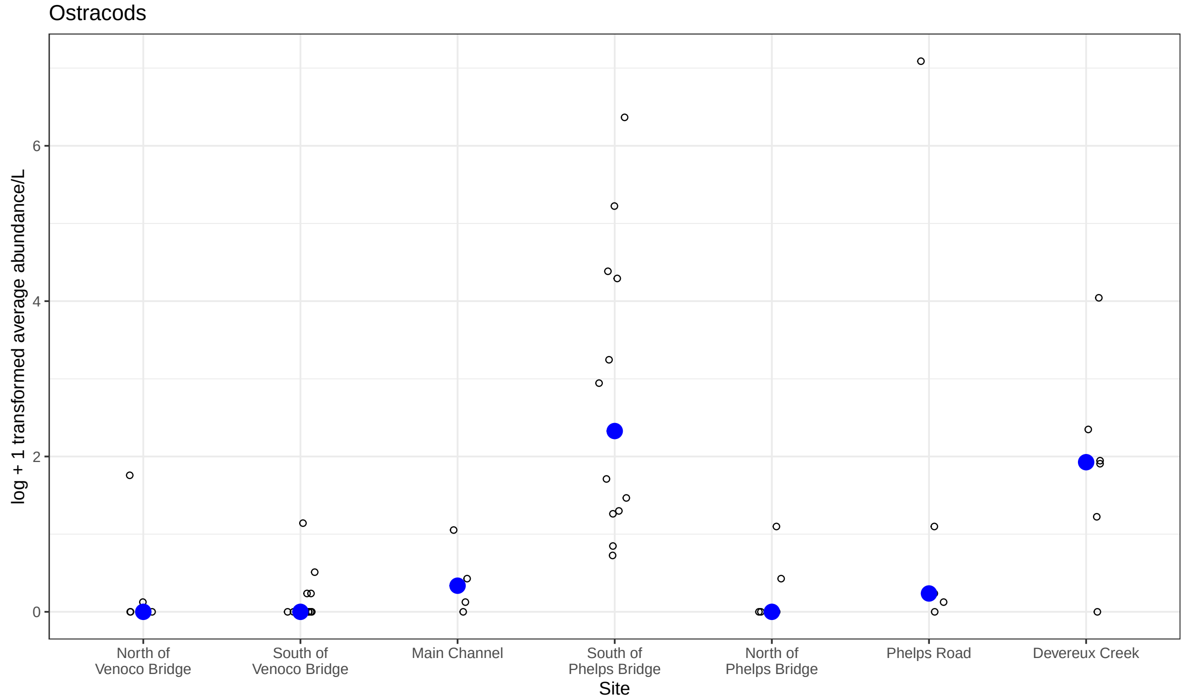
```
# create new dataframe from aq_ins_clean
ostracod_df <- aq_ins_clean |>
  # keep only ostracod observations
  filter(taxon == "ostracod") |>
  # create column with log transformed abundance
  mutate(log_abundance = log(ave_lit + 1))

# display 10 random rows from data frame
ostracod_df |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-02-12_NEC  2023-02-12 00:00:00 NEC   ostr~   2.13   NA      NA     NA
 2 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA      NA     NA
 3 2022-02-23_NVBR 2022-02-23 00:00:00 NVBR  ostr~   0       8.86   45.2   10.2
 4 2023-09-12_NVBR 2023-09-12 00:00:00 NVBR  ostr~   0       9.14   NA     20.3
 5 2022-08-15_NMC  2022-08-15 00:00:00 NMC   ostr~   0.133   7.61   NA     NA
 6 2022-02-21_NPB  2022-02-21 00:00:00 NPB   ostr~   1.33    7.39   34.1    0.34
 7 2022-08-12_NEC  2022-08-12 00:00:00 NEC   ostr~   0       7.37   NA     NA
 8 2023-05-04_NEC  2023-05-04 00:00:00 NEC   ostr~   0.667   8.05   NA      8.55
 9 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA      47     14.0
10 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71    1.98   9.59
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
```

```
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_abundance <dbl>
```

c. Coding

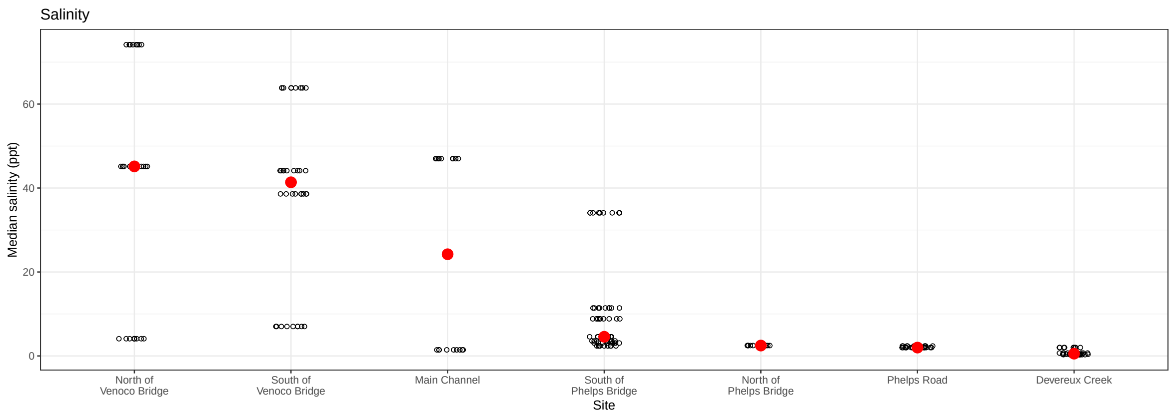
```
# base layer of ggplot as the ostracod dataframe we just created
ostracods <- ggplot(data = ostracod_df,
                    # x-axis: site
                    # y-axis: mean abundance/L
                    mapping = aes(x = site_full,
                                  y = log_abundance)) +
  # points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracods
```

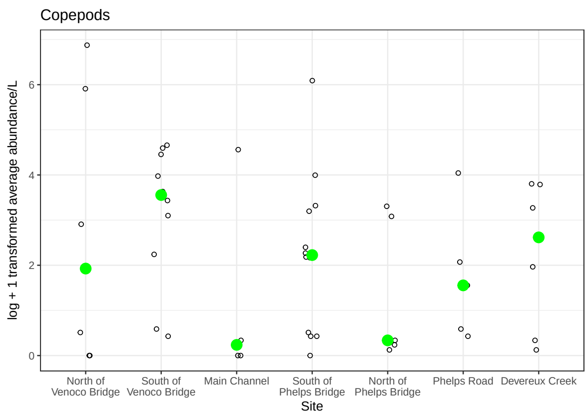


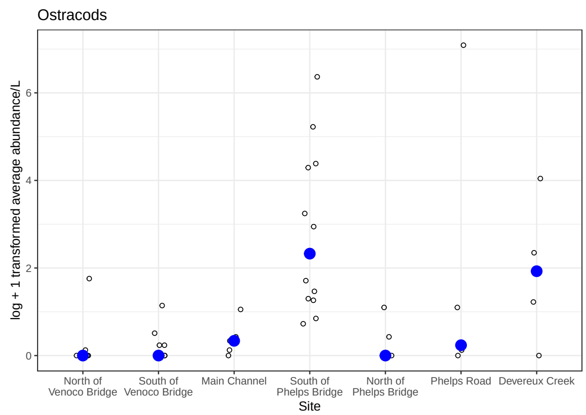
d. Creating multipanel figure

```
# base layer for salinity plot
salinity <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = site_full,
                                 y = med_sal)) +
  # points representing salinity observations
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # points representing median salinity
  stat_summary(fun = median,
               geom = "point",
               size = 4,
               color = "red") +
  # wrapping x-axis labels
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling axes and adding title
  labs(x = "Site",
       y = "Median salinity (ppt)",
```

```r
      title = "Salinity") +
  # changing theme
  theme_bw()

# create copepod dataframe
copepod_df <- aq_ins_clean |>
  # keep only copepod observations
  filter(taxon == "copepod") |>
  # create log transformed abundance column
  mutate(log_abundance = log(ave_lit + 1))

# base layer of copepod plot
copepods <- ggplot(data = copepod_df,
                   mapping = aes(x = site_full,
                                 y = log_abundance)) +
  # points representing observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # points representing medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "green") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling axes and adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # changing theme
  theme_bw()

# combining plots into multipanel figure
salinity / (copepods | ostracods)
```







e. Interpretation

As shown in the low position along the y axis of the blue and green median abundance values, both Ostracods and Copepods appear to be less abundant at Main Channel and North of Phelps Bridge. As shown in the high position along the y axis of the blue and green median abundance values, the highest median Ostracod abundance occurs South of Phelps Bridge, while the highest median Copepod abundance occurs South of Venoco Bridge.

While other factors may explain Ostracod abundance North of Phelps Bridge and at Phelps Road where salinity and abundance are both low, salinity appears to have an inverse relationship with Ostracod abundance across the rest of the sites. This is shown, for example, South of Phelps Bridge, where the red median salinity point is low on its y axis, and the blue median Ostracod abundance is high on its y axis.

2. Visualizing total abundance as a function of salinity

a. Planning

- x-axis: `med_sal` salinity (ppt)
- y-axis: `ave_lit` log-transformed average abundance/L
- color: `taxon`
- geometry: points
- facets: `facet_wrap(~ taxon, scales = "free")`

b. Wrangling

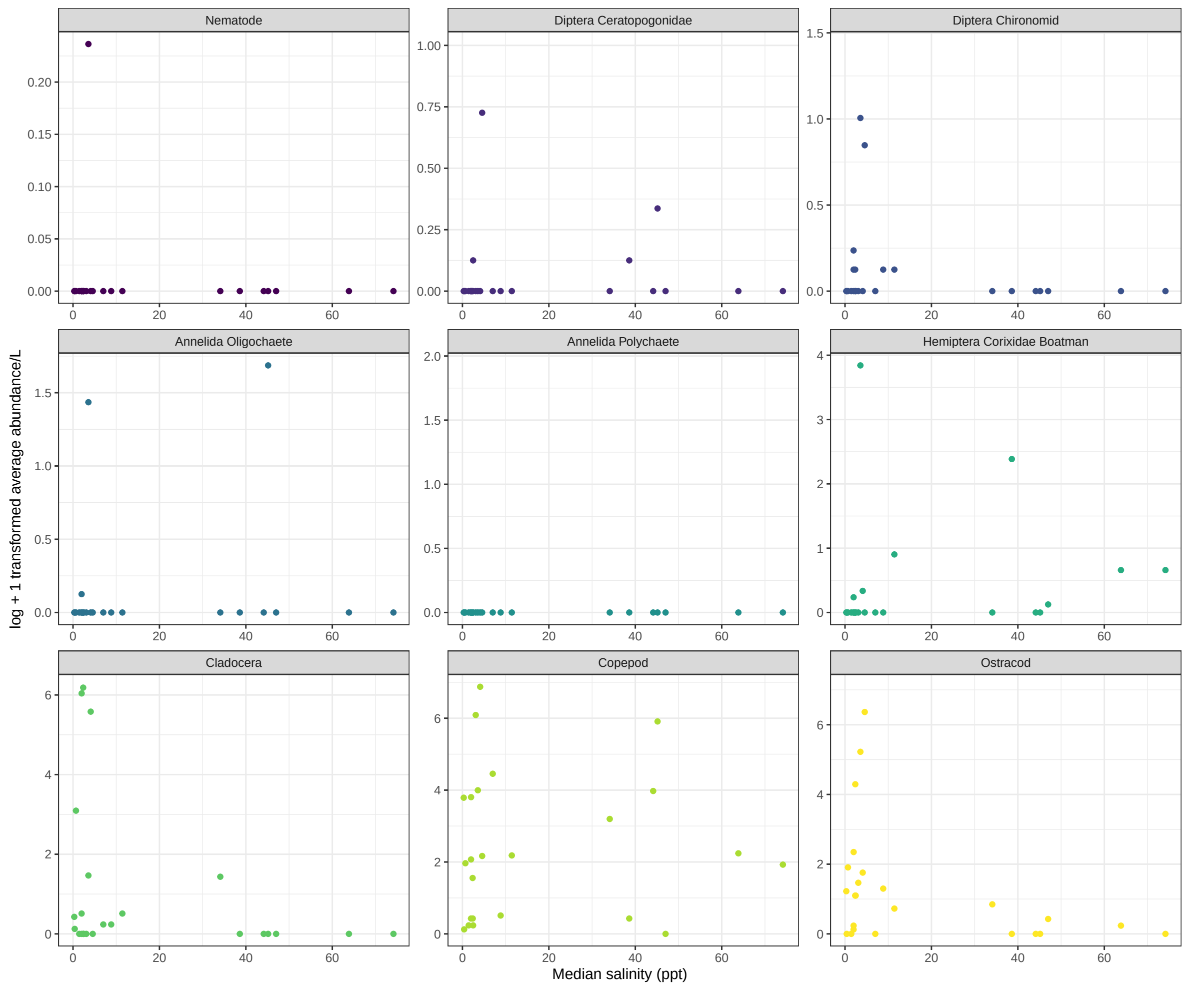
```
# create new dataframe from aq_ins_clean
salinity_abundance_df <- aq_ins_clean |>
  # keep only columns needed for visualization
  select(taxon,
         ave_lit,
         med_sal) |>
  # create new columns for plotting
  mutate(log_abundance = log(ave_lit + 1),
         # making sure each taxon is in title case
         taxon = to_any_case(taxon, case = "title")) |>
  # order taxon levels by maximum abundance
  mutate(taxon = fct_reorder(taxon,
                             ave_lit,
                             .fun = max))

# display 10 random rows from data frame
salinity_abundance_df |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   taxon                       ave_lit med_sal log_abundance
   <fct>                         <dbl>   <dbl>         <dbl>
 1 Annelida Oligochaete          0       2.36              0
 2 Hemiptera Corixidae Boatman   0       0.688             0
 3 Annelida Oligochaete          0      NA                 0
 4 Annelida Oligochaete          0      NA                 0
```

```
 5 Copepod                        0      NA            0
 6 Hemiptera Corixidae Boatman    0.4     4.1          0.336
 7 Annelida Polychaete            0      NA            0
 8 Diptera Chironomid             0.267   1.98         0.236
 9 Diptera Ceratopogonidae        0       0.398        0
10 Hemiptera Corixidae Boatman    0      NA            0
```

c. Code

```
# base layer as the salinity abundance dataframe we just created
salinity_abundance <- ggplot(data = salinity_abundance_df,
                             # x-axis: salinity
                             # y-axis: log transformed abundance
                             mapping = aes(x = med_sal,
                                           y = log_abundance,
                                           color = taxon)) +
  # points representing observations
  geom_point() +
  # create separate panels for each taxon
  facet_wrap(~ taxon,
             scales = "free") +
  # relabelling axes
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # changing color palette
  scale_color_viridis_d() +
  # removing legend
  theme_bw() +
  theme(legend.position = "none")

# displaying plot
salinity_abundance
```



d. Interpretation

Diptera Ceratopogonidae, Hemiptera Corixidae Boatman, Copepod, and Ostracod, appear to be more abundant at lower salinity, but also each experience another zone of elevated abundance, around 40 ppt salinity, where the points again increase in height along the y axis. Based on the uniformity of the position of the points along the y axis at all x axis values, it appears that Nematode and Annelida Polychaete abundance is not impacted by salinity. Based on the generally negative linear trend across most taxon, we can say that salinity limits aquatic invertebrate abundance at high levels (> 40 ppt) at NCOS.